

secure IT systems, cloud computing presents an added level of risk because:

- Services are outsourced to a third party
- It requires moving an application into hostile territory
- Multi-tenancy
- Loss of governance
- Legal and contractual risks

Cloud security is basically about three goals or objectives:

- Confidentiality (C)
- Integrity (I)
- Availability (A)

Cloud security and its compliance are the key components that are needed to protect cloud infrastructure against ever evolving new threats. It helps if organisations rationalise compliance requirements, controls, standards and best practices into centralised security policies administered consistently across virtual and physical infrastructures.

There are certain key areas where security levels are likely to be questionable and addressing them in an efficient manner is a critical task for easy and swift adoption of the cloud environment. Let's look at these areas more closely.

Identity and access management (IAM)

Unauthorised access to information resources in the cloud is a primary concern. In the current state of IAM technology, standards supported by cloud services providers offering various cloud service models (such as SaaS, PaaS and IaaS) are not consistent across providers. Google, Microsoft and Salesforce.com seem to demonstrate basic IAM capabilities. IAM should include the following: identity provisioning and de-provisioning, information privacy, linking, mapping, federation, attributes federation, authentication, SSO and authorisation.

Data security

Confidentiality, integrity and availability (CIA) form the golden trio of data security. Data security becomes more important while using cloud computing for all service models: IaaS, PaaS, and SaaS. Cloud environments are shared with multiple tenants and cloud service providers have privileged access to the data in those environments. Sensitive data stored in a cloud environment must be secured using a combination of strong encryption, access control, contractual liability, etc.

Network security

At the network level of infrastructure security, it is critical to distinguish between public clouds and private clouds. With private clouds, there may not be new attacks, vulnerabilities or changes in risk that information security personnel need to consider. In public clouds, changing security requirements will warrant more attention considering that it is the cloud service provider that is in control of resources.



Figure 1: Cloud security concerns



Figure 2: C-I-A

Virtualisation security

Virtualisation technologies enable multi-tenancy in cloud business models by providing a scalable and shared resource platform for all tenants. The threat of an unauthorised virtual machine (VM) is far higher because it is pretty easy to create and deploy virtual machines. Organisations need to enforce strict policies on the use of such virtual machine environments. VM images can be copied, along with the data and applications that they hold. These images can be brought back online to an unsecured network. It becomes easier for an attacker to access the contents managed within the copied image. Deployment of virtualisation in any organisation should be a controlled and policy-driven roll-out like any other computer platform, software or application.

Physical security

Cloud resources such as servers, routers, storage devices and power supplies that support cloud operations should be physically secure. Safeguards include the adequate control and monitoring of physical access using biometric access control measures and closed circuit television (CCTV) monitoring. Cloud service providers need to clearly explain how they manage physical access to the servers that host client workloads and support client data.

Use cases

Application firewall to protect apps in the private cloud

Let's look at a hypothetical case. The XYZ Market is a new shopping channel led by two seasoned industrialists. They have a private cloud for their organisation. XYZ Market's HR and business departments have applications named 'Time and Expense management' and 'E-commerce' respectively, which they want to deploy on a private cloud environment to gain agility, high availability and fault tolerance. Each application has two tiers – Web and the data base. The main concern of both departments is how to keep both applications secure in a private cloud environment.

The departments need to enforce the following separation between the applications and tiers of each application using the networking and security firewall